ORIGINAL ARTICLE

Breast Cancer in the Middle Eastern Population of California, 1988–2004

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■ **Abstract:** This report presents the patterns of incidence, survival, and mortality of breast cancer in the Middle Eastern (ME) population of California. Cases were identified through surname recognition and population estimates were obtained from census public use files. Rates, trends, and survival in this ethnic group were compared with the non-Hispanic White (NHW) of California, as well as natives in the Middle East. Age-adjusted incidence rates for the *insitu* (22.8), invasive (126.2), and mortality (23.2) in ME women were significantly lower than similar rates of 26.0, 146.9, and 30.6 in the NHW women. Incidence rate in ME women in California was higher than rates in women in the Middle East. Lower rates for early stage and higher rates for late stage diagnoses in this ethnic population suggest lack of optimal access to preventive healthcare. Relative survival in the two groups is negatively associated with stage at diagnosis and is slightly higher in ME women, probably due to large numbers of lost to follow-up in ME women suggesting the presence of salmon bias. Positive association with socioeconomic standing was detected only in the NHW women. Incidence of breast cancer in ME men was significantly higher than that of NHW men. ■

Key Words: breast cancer, California, California Cancer Registry, Middle Eastern, salmon bias, survival

Breast cancer is the most frequently diagnosed malignancy in women in the United States (1), Europe (2), and most other parts of the world (3). It is, however, not evenly distributed among countries, regions, and ethnic populations (4). In the United States, white women have the highest incidence rate, and significant variations are reported in rates for other racial and ethnic populations (5,6). Breast cancer is basically a hormonally induced malignancy and some of the observed differences are explainable by ethnic differences in prevalence of such risk factors as age at menarche, age at first birth, number of children, and postmenopausal use of hormone replacement therapy.

Migrant studies have provided a fairly clear description of the dynamics of breast cancer and have shown how the risk of diagnosis is associated with age at the time of migration, duration of stay, and acculturation in the diet and ways of life in the host country (7). "Middle Eastern" (ME) is a distinct ethnicity that includes individuals with origin in an area that roughly extends from the Mediterranean Sea to the Indian subcontinent in the east, and includes North Africa and the Arabian Peninsula. This area covers a mix of Arabs and non-Arabs in Afghanistan, Armenia, Iran, Israel, and Turkey. Internationally, age-adjusted data indicate that incidence rates per 100,000 (AAIR) for female breast cancer in this area varies from 12.7 in Oman to 49.6 in Egypt and are significantly lower than the AAIR of 97.2 in the United States (8–13). Migrant studies of ME population in Australia (14), the Netherlands (15), Sweden (16), Germany (17), and Canada (18) have confirmed that incidence rates in these immigrants are in between the rates in their home and the host countries.

In the United States, ME population is rapidly increasing and is estimated to pass 2.5 million by 2010 (19). The highest concentration of this emerging ethnic population is in California with a total that was estimated at 800,000 in 2004 (20,21). Published information on this emerging ethnicity is rare and points to some significant differences in cancer rates between them and the general population (22). The main reason for the rarity of information is the lack of official recognition of this ethnicity. Census data combines them with the white race (23), and cancer registry data-bases do not have a separate ethnic

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grouping for them (24). Name recognition that is widely used for identification of other ethnicities can also be used as an alternative method for identification of this ethnic population.

wThe objective of this report is to present the patterns of incidence, survival, and mortality of breast cancer in ME population, as identified by name recognition, and compare it with the non-Hispanic White (NHW) in California. Comparisons are also made with the available rates from countries in the Middle East.

MATERIALS AND METHODS

Cases

All cases of breast cancer registered with the California Cancer Registry (CCR) from 1988 through 2004 were selected for this study. CCR is a well-described (25) population-based, follow-up registry that began statewide data collection in 1988.

Deaths

All death certificates issued for California residents between January 1988 and December 2004 that had breast cancer as underlying cause of death were selected for this study. Access to the electronic files of California death certificates was provided through the CCR.

Race/Ethnicity

Middle Eastern identification was achieved by matching a newly developed "Middle Eastern Surname List" (26) with the incidence and mortality files. Briefly, this list includes 49,610 common ME surnames and has a sensitivity of 89% and specificity of over 99% for identifying individuals with the ME origin. Matching of this surname list with the incidence and death certificate files resulted in identification of 5,090 cases of invasive, 923 cases of *insitu* breast cancers, and 922 breast cancer deaths with ME origin. As 95% of ME cases were also identified as NHW, they were all combined as one group in a mutually exclusive classification with other race/-ethnicity groups.

Socioeconomic status

The socioeconomic status (SES) of cases was determined according to a previously reported area based classification at the level of census block group (27). The SES of cases diagnosed prior to 1995 were based on 1990 census and for those diagnosed since 1995 was based on the 2000 census.

Population estimates

Official estimates for ME population were extracted from the Census Public Use Microdata Samples (PUMS) for 1990, and the American Community Survey (ACS) PUMS for 2000 through 2004 (21). The following ancestries were selected to represent the Middle East: Afghan, Algerian, Arabian, Armenian, Assyrian, Chaldean, Egyptian, Iranian, Iraqi, Israeli, Jordanian, Libyan, Middle Eastern, Moroccan, Palestinian, Syrian, Tunisian, Turkish, and Yemeni. Population estimates for individual years from 1988 through 1999 were calculated by assuming simple linear extrapolation based on the 1990 census and 2000 ACS PUMS. Estimates for 2001 through 2004 were directly obtained from the ACS PUMS. Population estimates for the NHW was obtained through the CCR from the California Department of Finance (28) and were further modified by subtracting the ME population form it.

Middle East rates

To estimate the effects of migration, cancer rates for countries in the Middle East were obtained from the Globocan 2002 (29). The accuracy of the data presented in Globocan varies by country from national reports to various types of statistical estimates based on the best available data. To obtain a combined incidence rate for the Middle East, the following countries were grouped: Afghanistan, Algeria, Armenia, Azerbaijan, Bahrain, Egypt, Georgia, Iran, Iraq, Israel, Jordan, Kuwait, Lebanon, Libya, Morocco, Oman, Qatar, Saudi-Arabia, Syria, Tunisia, Turkey, United Arab Emirates, and Yemen.

Statistical analysis

Incidence and mortality rates were age adjusted to the United States 2000 standard population and expressed per 100,000. For comparison with ME countries, two sets of rates were calculated. One was the average 1996–2001 that was adjusted to the World standard with 18 five-year age categories, and the other was the average 2001–2003 that was adjusted to the world standard population with five broad age categories (0–14, 15–44, 45–54, 55–64, and 65+). Survival analysis was based on cases with single primary that were diagnosed between 1988 and 2004 and were followed up until December 2004.

SEER*Stat (30) and SAS (31) were used to calculate age-adjusted rates and relative survival, and Joinpoint (32) was used for trend analysis. Rates were directly compared with the confidence limits set at 95% (33).

RESULTS

Table 1 presents the overall rates for breast cancer incidence and mortality in California from 1988 through 2004 by sex, behavior, and race/ethnicity. A total of 6,013 cases and 902 deaths (in women only) that were attributed to breast cancer were identified as ME and were compared with 287,412 cases and 53,087 deaths in the NHW population in California. The incidence rate ratio for *insitu* and invasive breast cancer and breast cancer mortality in ME women are 0.88 (22.8/26.0, 95% CI: 0.82, 0.93), 0.86 (126.2/146.9, 95% CI: 0.84, 0.88), and 0.76 (23.2/30.6, 95% CI: 0.70, 0.81), respectively. In ME men, the incidence rate ratio for all breast cancers is 1.40 (1.80/1.29, 95% CI: 1.05, 1.85).

Figure 1 represents the regression analysis of the time trends for female breast cancer incidence by ethnicity and behavior. Incidence trends in the two groups follow similar patterns with slightly different magnitude. The rapidly increasing incidence of *insitu* cases in NHW women has leveled since 2000, while it is still raising in ME women. Trends for invasive cases changes to a decline in 2000–2001 for both ethnic women. Mortality trends in both groups follow a same pattern of steady decline (not shown).

Table 2 presents the results of segmented regression trend analysis for the ME and NHW populations. As shown, patterns of time trends are quite similar in the two ethnic groups with some variations in the magnitude. The only notable difference is in the pattern for *insitu* cases in ME women that continue to rise.

Figure 2 presents the trend in incidence of male breast cancer by ethnicity. Data presented in this figure combines the *insitu* and invasive cases. Apart from wide fluctuations in rates, the 5-year moving average curves clearly show that the incidence of male breast

Table 1. Breast Cancer Incidence and Mortality by Sex, Behavior, and Race/Ethnicity California 1988–2004

	Male				Female					
Race/ethnicity	Cases	Rate	Deaths	Rate	*Insitu*	Rate	Invasive	Rate	Deaths	Rate
All races combined	2,384	1.17*	512	0.26	57,782	22.75	332,855	129.32	71,473	27.68
Middle Eastern	73	1.80	18	0.49	915	22.76	5,025	126.16	902	23.25
Non-Hispanic White	1,780	1.29*	384	0.29	41,662	26.00*	243,970	146.90*	53,087	30.59*
Non-Hispanic Black	191	1.66	47	0.44	3,054	19.69	18,906	120.69	5,677	36.86*
Hispanic	190	0.66*	43	0.18*	6,133	13.90*	39,297	88.56*	7,864	18.77*
Non-Hispanic Asian/PI	111	0.57*	19	0.11*	5,039	17.98*	22,929	82.16*	3,766	14.08*

PI, Pacific Islander. *Statistically different from rates in the Middle Eastern group at $p < 0.05$. Rates are age adjusted to the US 2000 population standards. Race/ethnicity classification is mutually exclusive and Middle Eastern cases are excluded from all other race/ethnicities. All races combined include other and unknown races.

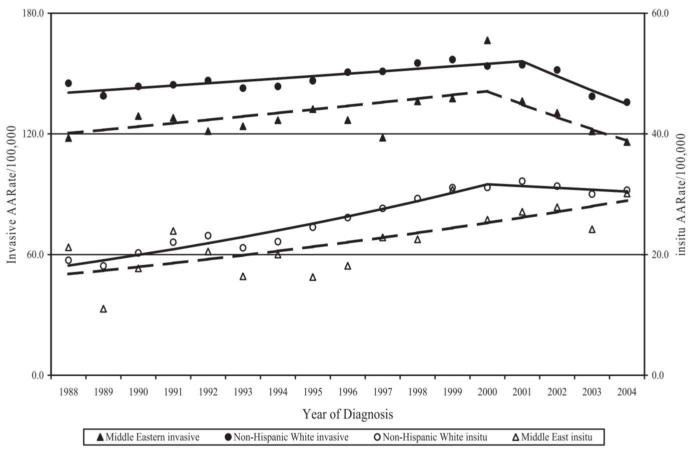

Figure 1. Trends in incidence of female breast cancer by ethnicity and behavior, California 1988–2004.

Table 2. Trends in Female Breast Cancer Incidence and Mortality By Race/Ethnicity, California 1988–2004

	Period	Beginning	Ending	APC	PC	p-value
Insitu						
Middle Eastern	1988–2004	21.21	30.13	3.48	42.06	0.001
Non-Hispanic White	1988–2000	19.05	31.13	4.72	63.41	0.000
	2000–2004	31.13	30.66	−0.96	−1.51	0.576
INVASIVE						
Middle Eastern	1988–2000	117.97	166.50	1.34	17.95	0.010
	2000–2004	166.50	115.95	−4.70	−21.65	0.020
Non-Hispanic White	1988–2001	145.16	154.33	6.32	0.81	0.008
	2001–2004	154.33	135.74	−12.05	−4.76	0.049
Mortality						
Middle Eastern	1988–2004	29.38	16.98	−3.09	−42.21	0.003
Non-Hispanic White	1988–2004	35.08	25.76	−2.13	−26.57	0.000

PC, percent change in the period; APC, annual percent change.

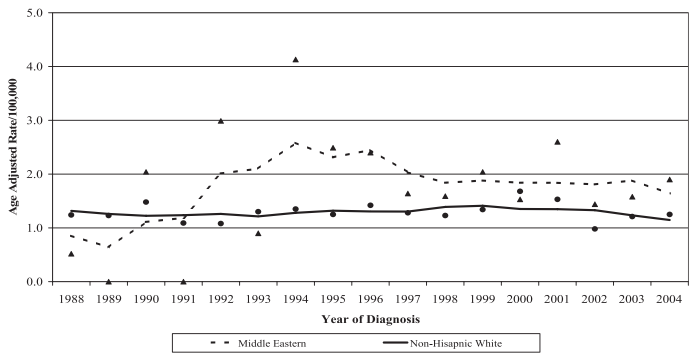

Figure 2. Trends (line based on 5-year moving average) in incidence of male breast cancer by ethnicity, California 1988–2004.

cancer in ME men is noticeably higher than that of the NHW men.

The 5-year relative survivals by ethnicity and stage at diagnosis are presented in Table 3. Overall, the relative survival of 100% for the *insitu* and stage I cases is reduced to less than 20% for stage IV cases with distant metastasis. Relative survivals in ME women are slightly better than that of the NHW, although in most stages, the differences are statistically not significant, except for stage IIA. This may be due to the lost to follow-up cases that are significantly higher in ME women and is positively associated with stage at diagnosis.

Table 3. Five-Year Relative Survival (%) and Percent LTFU Female Breast Cancer by Race⁄Ethnicity and Stage at Diagnosis, California 1988–1999

AJCC Stage	Non-Hispanic White Cases	Non-Hispanic White Survival	Non-Hispanic White LTFU	Middle Eastern Cases	Middle Eastern Survival	Middle Eastern LTFU
Insitu	16,415	100.00	0.85	339	100.00	3.54
Stage I	49,625	100.00	0.56	760	100.00	1.32
Stage IIA	27,921	90.87*	0.58	594	93.66	3.37
Stage IIB	14,215	76.96	0.53	375	79.71	4.53
Stage IIIA	3,107	60.33	0.51	83	70.19	6.02
Stage IIIB	3,464	45.45	0.49	94	43.83	7.45
Stage IV	5,124	16.97	0.27	120	19.54	5.00

AJCC, American Joint Committee on Cancer; LTFU, lost to follow-up. Relative survival is based on cases with single primary with follow-up information as of December 2004. *Statistically significant from survival in Middle Eastern group at p < 0.05.

Table 4 presents the impact of SES on relative survival by stage. While there is no association with SES for early stage cancers in either of the ethnic groups, higher SES is associated with better survival in more advanced cases only in NHW women.

Table 5 presents female breast cancer rates in NHW and ME populations of California as well as rates for selected countries in the Middle East. The comparative data come from two different sources, however the pattern is similar. Incidence of invasive breast cancer in ME women in California is closer to NHW women in California than to women in the home countries.

Table 4. Five-Year Relative Survival (%) for Female Breast Cancer by Stage at Diagnosis, Race/Ethnicity, and SES, California 1988–1999

	Low SES		High SES		
	Cases	Relative	Cases	Relative	*Z*-value
Middle Eastern					
Insitu	23	100.00	149	100.00	0.000
Stage I	40	99.18	314	100.00	0.000
Stage IIA	38	100.00	208	96.70	−0.628
Stage IIB	32	85.20	119	80.29	−0.519
Stage IIIA	7	62.27	32	83.67	0.966
Stage IIIB	7	60.36	24	43.34	−0.619
Stage IV	25	17.76	32	26.19	0.709
Non-Hispanic White					
Insitu	940	100.00	5,885	100.00	0.000
Stage I	3,226	100.00	15,996	100.00	0.000
Stage IIA	2,167	89.81	8,541	93.10	2.889
Stage IIB	1,157	73.21	4,259	80.45	4.094
Stage IIIA	246	52.08	898	63.27	2.783
Stage IIIB	346	39.19	854	54.83	4.063
Stage IV	558	12.16	1,255	20.65	3.931

SES, socioeconomic status. Relative survival is based on cases with single primary and follow-up information

Table 5. Female Breast Cancer by Country of Origin

Country	Rate	Count
Middle East (2002)[1]	27.5	46,033
Middle Eastern in California (2001–2003)[2]	99.0	1,212
Non-Hispanic White in California (2001–2003)[2]	111.8	45,139
Non-Hispanic White in California (1996–2001)[3]	111.0	91,095
Middle Eastern in California (1996–2001)[3]	98.8	2,020
US SEER (1999–2001)[4]	97.2	78,802
Cyprus (1998–2001)[4]	57.7	1,066
Israel Arabs (1996–2001)[4]	36.7	762
Egypt (1999–2001)[4]	49.6	1,945
Jordan (1996–2001)[4]	38.0	2,930

[1]Globocan 2002; [2]Age adjusted to world Standard population (four categories); [3]Age adjusted to World Standard Million (18 categories); [4]Middle East Cancer Consortium 2006.

DISCUSSION

This is probably the first report of breast cancer in ME population in California and presents few noteworthy results. Generally, patterns of female breast cancer in ME women is similar to that of the NHW women, with a relative risk of 0.85 that is much higher than 0.25 that is reported for ME women in most European studies (15). One of the probable reasons for this observation is that about 30% of ME population in this study are second generation immigrants who are born in the United States, are highly acculturated, and are expected to have higher incidence rate than their parents (7). The proportion of ME women diagnosed at the *insitu* and stage I that are usually detected through screening is 0.88, while it is 1.27 for more advanced cases with clinical symptoms. This pattern is often noticed among recent immigrants who have limited access to preventive healthcare services.

Pattern of the 5-year relative survival in ME women is similar to that of NHW women with two distinct differences. First, no statistical association with SES is detected in them. This may be due to the small number of cases. Second, the relative survivals by more advanced stages in ME women is slightly better than that of the NHW women. This observation, although statistically not significant, is probably associated with the higher number of lost to follow-up cases in this group, representing cases may have returned home to a more familiar surroundings following diagnosis with advanced stages. This phenomena that is generally known as ‘‘salmon bias’’ has previously been discussed as an explanation for lower mortality rates among the Hispanic population in the United States (34) and Mediterranean immigrants in Europe (35,36), but it is probably the first time that is directly reported in a survival analysis.

Middle Eastern women who are living in California have significantly higher incidence of breast cancer compared with women in their home countries. There are few possible reasons for this observation. One reason is the presence of US born ME women in this study. These women more closely resemble American women than those who live in the Middle East. The second reason is that early detection of subclinical cases and data collection in California is much better than most of the ME countries and the observed difference may partially represent better methodology. Another reason may reside in the fact that migrant populations are generally not a true representative of their countries and may have disease patterns that are not comparable with the population at home. In this study, a sizeable proportion of cases with places of birth in Iran, Turkey, and Afghanistan were non-Muslims, while over 90% of the population in these countries are Muslims. A separate and significant observation in this study is that the incidence of breast cancer in ME men is significantly higher than NHW men. Reasons for this are not clear, although genetic predisposition may play a significant part.

In conclusion, this report describes the epidemiologic patterns of breast cancer in ME population of California. Some of the findings are noteworthy and can be used for further research on the issue of breast cancer in this ethnic population that has up to now been hidden in the White race.

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